Gabarito Questoes Computacionais

Tuffy

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Lista 3 a 11 - Econometria I - 2026

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Lista 3

Questao 5)

Vamos comecar abrindo a base de dados solicitada pelo exercicio.

```r
library(wooldridge)
library(stargazer)

data("hprice1")
head(hprice1)
```

```
##     price assess bdrms lotsize sqrft colonial   lprice  lassess llotsize
## 1 300.000  349.1     4    6126  2438        1 5.703783 5.855359 8.720297
## 2 370.000  351.5     3    9903  2076        1 5.913503 5.862210 9.200593
## 3 191.000  217.7     3    5200  1374        0 5.252274 5.383118 8.556414
## 4 195.000  231.8     3    4600  1448        1 5.273000 5.445875 8.433811
## 5 373.000  319.1     4    6095  2514        1 5.921578 5.765504 8.715224
## 6 466.275  414.5     5    8566  2754        1 6.144775 6.027073 9.055556
##     lsqrft
## 1 7.798934
## 2 7.638198
## 3 7.225482
## 4 7.277938
## 5 7.829630
## 6 7.920810
```

- **(a)**

Vamos estimar por MQO a regressao pedida:

```r
mod3 <- lm(price ~ sqrft + bdrms, data = hprice1)
summary(mod3)
```

```
##
## Call:
## lm(formula = price ~ sqrft + bdrms, data = hprice1)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -127.627  -42.876   -7.051   32.589  229.003
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -19.31500   31.04662  -0.622    0.536
## sqrft         0.12844    0.01382   9.291 1.39e-14 ***
## bdrms        15.19819    9.48352   1.603    0.113
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 63.04 on 85 degrees of freedom
## Multiple R-squared:  0.6319, Adjusted R-squared:  0.6233
## F-statistic: 72.96 on 2 and 85 DF,  p-value: < 2.2e-16
```

```r
stargazer(mod3, type = "text", digits = 4)
```

```
##
## ===============================================
##                         Dependent variable:
##                     ---------------------------
##                                price
## -----------------------------------------------
## sqrft                        0.1284***
##                              (0.0138)
##
## bdrms                         15.1982
##                              (9.4835)
##
## Constant                     -19.3150
##                              (31.0466)
##
## -----------------------------------------------
## Observations                    88
## R2                            0.6319
## Adjusted R2                   0.6233
## Residual Std. Error      63.0448 (df = 85)
## F Statistic           72.9635*** (df = 2; 85)
## ===============================================
## Note:               *p<0.1; **p<0.05; ***p<0.01
```

```r
coef(mod3)
```

```
## (Intercept)       sqrft       bdrms
## -19.3149958   0.1284362  15.1981910
```

Assim, a equacao estimada fica:

$$\widehat{price} = \hat{\beta}_0 + \hat{\beta}_1\, sqrft + \hat{\beta}_2\, bdrms$$

com os coeficientes numericos dados na saida acima.

- **(b)**

O efeito de um quarto adicional, mantendo a area fixa, e dado pelo coeficiente de `bdrms`.

```r
coef(mod3)["bdrms"]
```

```
##    bdrms
## 15.19819
```

Ou seja, um quarto a mais aumenta o preco previsto em 15.198 mil dolares, ceteris paribus.

- **(c)**

Se o quarto adicional vier acompanhado de 140 pes quadrados extras, o aumento previsto no preco sera:

```r
delta_price <- 140 * coef(mod3)["sqrft"] + coef(mod3)["bdrms"]
delta_price
```

```
##    sqrft
## 33.17926
```

Portanto, o aumento estimado e de 33.179 mil dolares.

- **(d)**

O percentual da variacao de `price` explicado por `sqrft` e `bdrms` e medido pelo R^2.

```r
summary(mod3)$r.squared
```

```
## [1] 0.6319184
```

Assim, o modelo explica aproximadamente 63.19% da variacao do preco.

- **(e)**

Para `sqrft = 2438` e `bdrms = 4`, a previsao e:

```r
nova_casa <- data.frame(sqrft = 2438, bdrms = 4)
predict(mod3, newdata = nova_casa)
```

```
##        1
## 354.6052
```

- **(f)**

O residuo da primeira observacao e o preco observado menos o preco previsto.

```r
preco_previsto_1 <- fitted(mod3)[1]
preco_observado_1 <- hprice1$price[1]
residuo_1 <- resid(mod3)[1]

preco_previsto_1
```

```
##        1
## 354.6052
```

```r
preco_observado_1
```

```
## [1] 300
```

```r
residuo_1
```

```
##         1
## -54.60525
```

Se o residuo for negativo, o modelo superestimou o preco da casa. Se for positivo, o modelo subestimou.

Lista 4

Questao 5)

Vamos carregar novamente os pacotes e a base usada no exercicio.

```r
library(wooldridge)
library(stargazer)

options(scipen = 999)
data("discrim")
head(discrim)
```

```
##   psoda pfries pentree wagest nmgrs nregs hrsopen  emp psoda2 pfries2 pentree2
## 1  1.12   1.06    1.02   4.25     3     5    16.0 27.5   1.11    1.11     1.05
## 2  1.06   0.91    0.95   4.75     3     3    16.5 21.5   1.05    0.89     0.95
## 3  1.06   0.91    0.98   4.25     3     5    18.0 30.0   1.05    0.94     0.98
## 4  1.12   1.02    1.06   5.00     4     5    16.0 27.5   1.15    1.05     1.05
```

```
## 5  1.12     NA    0.49   5.00     3     3    16.0  5.0   1.04    1.01     0.58
## 6  1.06   0.95    1.01   4.25     4     4    15.0 17.5   1.05    0.94     1.00
##   wagest2 nmgrs2 nregs2 hrsopen2 emp2 compown chain density    crmrte state
## 1    5.05      5      5     15.0 27.0       1     3    4030 0.0528866     1
## 2    5.05      4      3     17.5 24.5       0     1    4030 0.0528866     1
## 3    5.05      4      5     17.5 25.0       0     1   11400 0.0360003     1
## 4    5.05      4      5     16.0   NA       0     3    8345 0.0484232     1
## 5    5.05      3      3     16.0 12.0       0     1     720 0.0615890     1
## 6    5.05      3      4     15.0 28.0       0     1    4424 0.0334823     1
##     prpblck    prppov   prpncar hseval nstores income county     lpsoda
## 1 0.1711542 0.0365789 0.0788428 148300       3  44534     18 0.11332869
## 2 0.1711542 0.0365789 0.0788428 148300       3  44534     18 0.05826885
## 3 0.0473602 0.0879072 0.2694298 169200       3  41164     12 0.05826885
## 4 0.0528394 0.0591227 0.1366903 171600       3  50366     10 0.11332869
## 5 0.0344800 0.0254145 0.0738020 249100       1  72287     10 0.11332869
## 6 0.0591327 0.0835001 0.1151341 148000       2  44515     18 0.05826885
##       lpfries  lhseval  lincome ldensity NJ BK KFC RR
## 1  0.05826885 11.90699 10.70401 8.301521  1  0   0  1
## 2 -0.09431065 11.90699 10.70401 8.301521  1  1   0  0
## 3 -0.09431065 12.03884 10.62532 9.341369  1  1   0  0
## 4  0.01980261 12.05292 10.82707 9.029418  1  0   0  1
## 5          NA 12.42561 11.18840 6.579251  1  1   0  0
## 6 -0.05129331 11.90497 10.70358 8.394799  1  1   0  0
```

- **(a)**

Primeiro, calculamos medias e desvios-padrao de `prpblck` e `income`.

```r
mean_prpblck <- mean(discrim$prpblck, na.rm = TRUE)
sd_prpblck <- sd(discrim$prpblck, na.rm = TRUE)

mean_income <- mean(discrim$income, na.rm = TRUE)
sd_income <- sd(discrim$income, na.rm = TRUE)

data.frame(
  variavel = c("prpblck", "income"),
  media = c(mean_prpblck, mean_income),
  desvio_padrao = c(sd_prpblck, sd_income)
)
```

```
##   variavel         media desvio_padrao
## 1  prpblck     0.1134864     0.1824165
## 2   income 47053.7848411 13179.2860689
```

`prpblck` e uma proporcao, variando entre 0 e 1. `income` mede renda familiar mediana em dolares.

- **(b)**

O enunciado pede a regressao com `psoda`.

```r
mod4_b <- lm(psoda ~ prpblck + income, data = discrim)
summary(mod4_b)
```

```
##
## Call:
## lm(formula = psoda ~ prpblck + income, data = discrim)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.29401 -0.05242  0.00333  0.04231  0.44322
##
## Coefficients:
##                 Estimate   Std. Error t value             Pr(>|t|)
## (Intercept) 0.9563196258 0.0189920097  50.354 < 0.0000000000000002 ***
## prpblck     0.1149881907 0.0260006361   4.423            0.0000126 ***
## income      0.0000016027 0.0000003618   4.430            0.0000122 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.08611 on 398 degrees of freedom
##   (9 observations deleted due to missingness)
## Multiple R-squared:  0.06422,    Adjusted R-squared:  0.05952
## F-statistic: 13.66 on 2 and 398 DF,  p-value: 0.000001835
```

```r
stargazer(mod4_b, type = "text", digits = 4)
```

```
##
## ===============================================
##                         Dependent variable:
##                     ---------------------------
##                               psoda
## -----------------------------------------------
## prpblck                     0.1150***
##                             (0.0260)
##
## income                     0.000002***
##                            (0.000000)
##
## Constant                    0.9563***
##                             (0.0190)
##
## -----------------------------------------------
## Observations                  401
## R2                           0.0642
## Adjusted R2                  0.0595
## Residual Std. Error      0.0861 (df = 398)
## F Statistic          13.6569*** (df = 2; 398)
## ===============================================
## Note:               *p<0.1; **p<0.05; ***p<0.01
```

```r
coef(mod4_b)
```

```
##    (Intercept)        prpblck         income
## 0.956319625816 0.114988190739 0.000001602674
```

Como o modelo esta em nivel-nivel, o coeficiente de `prpblck` mede a variacao em dolares no preco do refrigerante para um aumento de uma unidade em `prpblck`. Como uma unidade equivale a 100 pontos percentuais, um aumento de 0,10 em `prpblck` implica variacao prevista de:

```r
efeito_10pp_psoda <- 0.10 * coef(mod4_b)["prpblck"]
efeito_10pp_psoda
```

```
##    prpblck
## 0.01149882
```

Ou seja, 0.0115 dolar por refrigerante. Isso permite discutir se o efeito e economicamente grande ou pequeno.

- **(c)**

Agora estimamos a regressao simples de `psoda` sobre `prpblck`.

```r
mod4_c <- lm(psoda ~ prpblck, data = discrim)
summary(mod4_c)
```

```
##
## Call:
## lm(formula = psoda ~ prpblck, data = discrim)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.30884 -0.05963  0.01135  0.03206  0.44840
##
## Coefficients:
##             Estimate Std. Error t value             Pr(>|t|)
## (Intercept)  1.03740    0.00519  199.87 < 0.0000000000000002 ***
## prpblck      0.06493    0.02396    2.71              0.00702 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.0881 on 399 degrees of freedom
##   (9 observations deleted due to missingness)
## Multiple R-squared:  0.01808,    Adjusted R-squared:  0.01561
## F-statistic: 7.345 on 1 and 399 DF,  p-value: 0.007015
```

```r
stargazer(mod4_c, type = "text", digits = 4)
```

```
##
## ===============================================
##                         Dependent variable:
##                     ---------------------------
##                                psoda
## -----------------------------------------------
## prpblck                     0.0649***
```

```
##                                (0.0240)
##
## Constant                       1.0374***
##                                (0.0052)
##
## ----------------------------------------------
## Observations                     401
## R2                              0.0181
## Adjusted R2                     0.0156
## Residual Std. Error      0.0881 (df = 399)
## F Statistic           7.3449*** (df = 1; 399)
## ==============================================
## Note:               *p<0.1; **p<0.05; ***p<0.01
```

```r
coef(mod4_c)
```

```
## (Intercept)     prpblck
##  1.03739915  0.06492687
```

Comparando o coeficiente de `prpblck` em (b) e (c), vemos como o controle por renda altera o efeito estimado.

- **(d)**

Agora estimamos o modelo semilog pedido pelo item:

```r
mod4_d <- lm(lpsoda ~ prpblck + lincome, data = discrim)
summary(mod4_d)
```

```
##
## Call:
## lm(formula = lpsoda ~ prpblck + lincome, data = discrim)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.33563 -0.04695  0.00658  0.04334  0.35413
##
## Coefficients:
##             Estimate Std. Error t value   Pr(>|t|)
## (Intercept) -0.79377    0.17943  -4.424 0.00001254 ***
## prpblck      0.12158    0.02575   4.722 0.00000324 ***
## lincome      0.07651    0.01660   4.610 0.00000543 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.0821 on 398 degrees of freedom
##   (9 observations deleted due to missingness)
## Multiple R-squared:  0.06809,    Adjusted R-squared:  0.06341
## F-statistic: 14.54 on 2 and 398 DF,  p-value: 0.0000008039
```

```r
stargazer(mod4_d, type = "text", digits = 4)
```

```
##
## ================================================
##                         Dependent variable:
##                     ---------------------------
##                              lpsoda
## ------------------------------------------------
## prpblck                     0.1216***
##                             (0.0257)
##
## lincome                     0.0765***
##                             (0.0166)
##
## Constant                   -0.7938***
##                             (0.1794)
##
## ------------------------------------------------
## Observations                   401
## R2                            0.0681
## Adjusted R2                   0.0634
## Residual Std. Error      0.0821 (df = 398)
## F Statistic          14.5405*** (df = 2; 398)
## ================================================
## Note:               *p<0.1; **p<0.05; ***p<0.01
```

```r
coef(mod4_d)
```

```
## (Intercept)     prpblck     lincome
## -0.79376804  0.12158026  0.07651138
```

Em um modelo log-nivel, a variacao percentual aproximada em `psoda` para um aumento de 0,22 em `prpblck` e:

```r
delta_pct <- 100 * coef(mod4_d)["prpblck"] * 0.22
delta_pct
```

```
##  prpblck
## 2.674766
```

Logo, a variacao percentual estimada e de aproximadamente 2.67%.

- **(e)**

Agora adicionamos `prppov` ao modelo do item anterior.

```r
mod4_e <- lm(lpsoda ~ prpblck + lincome + prppov, data = discrim)
summary(mod4_e)
```

```
##
## Call:
## lm(formula = lpsoda ~ prpblck + lincome + prppov, data = discrim)
##
```

```
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.32218 -0.04648  0.00651  0.04272  0.35622
##
## Coefficients:
##             Estimate Std. Error t value   Pr(>|t|)
## (Intercept) -1.46333    0.29371  -4.982 0.00000094 ***
## prpblck      0.07281    0.03068   2.373     0.0181 *
## lincome      0.13696    0.02676   5.119 0.00000048 ***
## prppov       0.38036    0.13279   2.864     0.0044 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.08137 on 397 degrees of freedom
##   (9 observations deleted due to missingness)
## Multiple R-squared:  0.08696,    Adjusted R-squared:  0.08006
## F-statistic:  12.6 on 3 and 397 DF,  p-value: 0.00000006917
```

```r
stargazer(mod4_e, type = "text", digits = 4)
```

```
##
## ===============================================
##                         Dependent variable:
##                     ---------------------------
##                              lpsoda
## -----------------------------------------------
## prpblck                     0.0728**
##                             (0.0307)
##
## lincome                     0.1370***
##                             (0.0268)
##
## prppov                      0.3804***
##                             (0.1328)
##
## Constant                   -1.4633***
##                             (0.2937)
##
## -----------------------------------------------
## Observations                   401
## R2                            0.0870
## Adjusted R2                   0.0801
## Residual Std. Error      0.0814 (df = 397)
## F Statistic          12.6040*** (df = 3; 397)
## ===============================================
## Note:               *p<0.1; **p<0.05; ***p<0.01
```

```r
coef(mod4_e)
```

```
## (Intercept)     prpblck     lincome      prppov
## -1.46333262  0.07280721  0.13695528  0.38035997
```

O ponto central e comparar os coeficientes de `prpblck` em `mod4_d` e `mod4_e`. Em geral, ao incluir `prppov`, a magnitude de `prpblck` cai, mostrando que parte do efeito anterior refletia diferencas de pobreza entre localidades.

- **(f)**

```
corr_lincome_prppov <- cor(discrim$lincome, discrim$prppov, use = "complete.obs")
corr_lincome_prppov
```

```
## [1] -0.838467
```

A correlacao e fortemente negativa, o que faz sentido: areas com maior renda tendem a ter menor proporcao de pessoas abaixo da linha de pobreza.

- **(g)**

A afirmacao esta incorreta. Colinearidade alta entre `lincome` e `prppov` pode inflar erros-padrao e reduzir a precisao das estimativas individuais, mas nao gera vies de MQO nem obriga a retirar uma das variaveis. Se ambas forem relevantes do ponto de vista teorico, podem permanecer no modelo.

Lista 5

Questao 5)

Agora fazemos o gabarito computacional da Lista 5 usando a base `DISCRIM`, exatamente como pedido no enunciado da lista.

```
library(wooldridge)
library(stargazer)

options(scipen = 999)
data("discrim")

head(discrim)
```

```
##   psoda pfries pentree wagest nmgrs nregs hrsopen  emp psoda2 pfries2 pentree2
## 1  1.12   1.06    1.02   4.25     3     5    16.0 27.5   1.11    1.11     1.05
## 2  1.06   0.91    0.95   4.75     3     3    16.5 21.5   1.05    0.89     0.95
## 3  1.06   0.91    0.98   4.25     3     5    18.0 30.0   1.05    0.94     0.98
## 4  1.12   1.02    1.06   5.00     4     5    16.0 27.5   1.15    1.05     1.05
## 5  1.12     NA    0.49   5.00     3     3    16.0  5.0   1.04    1.01     0.58
## 6  1.06   0.95    1.01   4.25     4     4    15.0 17.5   1.05    0.94     1.00
##   wagest2 nmgrs2 nregs2 hrsopen2 emp2 compown chain density    crmrte state
## 1    5.05      5      5     15.0 27.0       1     3    4030 0.0528866     1
## 2    5.05      4      3     17.5 24.5       0     1    4030 0.0528866     1
## 3    5.05      4      5     17.5 25.0       0     1   11400 0.0360003     1
## 4    5.05      4      5     16.0   NA       0     3    8345 0.0484232     1
## 5    5.05      3      3     16.0 12.0       0     1     720 0.0615890     1
## 6    5.05      3      4     15.0 28.0       0     1    4424 0.0334823     1
##     prpblck    prppov   prpncar hseval nstores income county     lpsoda
```

```
## 1 0.1711542 0.0365789 0.0788428 148300       3  44534     18 0.11332869
## 2 0.1711542 0.0365789 0.0788428 148300       3  44534     18 0.05826885
## 3 0.0473602 0.0879072 0.2694298 169200       3  41164     12 0.05826885
## 4 0.0528394 0.0591227 0.1366903 171600       3  50366     10 0.11332869
## 5 0.0344800 0.0254145 0.0738020 249100       1  72287     10 0.11332869
## 6 0.0591327 0.0835001 0.1151341 148000       2  44515     18 0.05826885
##       lpfries  lhseval  lincome ldensity NJ BK KFC RR
## 1  0.05826885 11.90699 10.70401 8.301521  1  0   0  1
## 2 -0.09431065 11.90699 10.70401 8.301521  1  1   0  0
## 3 -0.09431065 12.03884 10.62532 9.341369  1  1   0  0
## 4  0.01980261 12.05292 10.82707 9.029418  1  0   0  1
## 5          NA 12.42561 11.18840 6.579251  1  1   0  0
## 6 -0.05129331 11.90497 10.70358 8.394799  1  1   0  0
```

- **(a)**

Vamos estimar por MQO o modelo pedido:

```r
mod5_a <- lm(lpsoda ~ prpblck + lincome + prppov, data = discrim)
summary(mod5_a)
```

```
##
## Call:
## lm(formula = lpsoda ~ prpblck + lincome + prppov, data = discrim)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.32218 -0.04648  0.00651  0.04272  0.35622
##
## Coefficients:
##             Estimate Std. Error t value   Pr(>|t|)
## (Intercept) -1.46333    0.29371  -4.982 0.00000094 ***
## prpblck      0.07281    0.03068   2.373     0.0181 *
## lincome      0.13696    0.02676   5.119 0.00000048 ***
## prppov       0.38036    0.13279   2.864     0.0044 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.08137 on 397 degrees of freedom
##   (9 observations deleted due to missingness)
## Multiple R-squared:  0.08696,    Adjusted R-squared:  0.08006
## F-statistic:  12.6 on 3 and 397 DF,  p-value: 0.00000006917
```

```r
stargazer(mod5_a, type = "text", digits = 4)
```

```
##
## ===============================================
##                         Dependent variable:
##                     ---------------------------
##                              lpsoda
## -----------------------------------------------
## prpblck                     0.0728**
```

```
##                              (0.0307)
##
## lincome                      0.1370***
##                              (0.0268)
##
## prppov                       0.3804***
##                              (0.1328)
##
## Constant                    -1.4633***
##                              (0.2937)
##
## ----------------------------------------------
## Observations                    401
## R2                             0.0870
## Adjusted R2                    0.0801
## Residual Std. Error      0.0814 (df = 397)
## F Statistic          12.6040*** (df = 3; 397)
## ==============================================
## Note:               *p<0.1; **p<0.05; ***p<0.01
```

```r
coef(mod5_a)
```

```
## (Intercept)     prpblck     lincome      prppov
## -1.46333262  0.07280721  0.13695528  0.38035997
```

Com isso, a equacao estimada pode ser escrita como:

$$\widehat{\log(psoda)} = -1.463 + 0.0728\, prpblck + 0.137\, lincome + 0.380\, prppov$$

Temos $n = 401$ observacoes e $R^2 \approx 0.087$.

Agora vamos verificar se o intercepto e estatisticamente diferente de zero a 5% e a 1%.

```r
coef(summary(mod5_a))["(Intercept)", ]
```

```
##         Estimate       Std. Error          t value         Pr(>|t|)
## -1.4633326164663  0.2937110706789 -4.9822181134816  0.0000009400129
```

O p-valor bilateral do intercepto e aproximadamente 0.0000009. Portanto, rejeitamos $H_0 : \beta_0 = 0$ tanto a 5% quanto a 1%.

- **(b)**

Agora calculamos a correlacao entre `lincome` e `prppov` e registramos os p-valores das duas variaveis no modelo estimado em (a).

```r
cor(discrim$lincome, discrim$prppov, use = "complete.obs")
```

```
## [1] -0.838467
```

```r
coef(summary(mod5_a))[c("lincome", "prppov"), c("Estimate", "Std. Error", "Pr(>|t|)")]
```

```
##          Estimate Std. Error       Pr(>|t|)
## lincome 0.1369553 0.02675545 0.000000480201
## prppov  0.3803600 0.13279031 0.004400337780
```

A correlacao entre `lincome` e `prppov` e aproximadamente -0.8385, isto e, fortemente negativa. Mesmo assim, no modelo do item (a), ambas permanecem estatisticamente significantes:

- p-valor bilateral de `lincome`: 0.0000005;
- p-valor bilateral de `prppov`: 0.0044.
- **(c)**

O intercepto mede o valor previsto de $\log(psoda)$ quando `prpblck = 0`, `lincome = 0` e `prppov = 0`. Em particular, `lincome = 0` significaria `income = 1`, o que nao e um valor economicamente plausivel para a renda mediana de um CEP. Portanto, embora o intercepto seja estatisticamente diferente de zero, ele **nao tem interpretacao economica muito interessante** neste contexto.

- **(d)**

Agora calculamos o p-valor do teste

$$H_0 : \beta_2 = 1 \qquad \text{contra} \qquad H_1 : \beta_2 < 1,$$

em que β_2 e o coeficiente de `lincome`.

```r
beta2_hat <- coef(mod5_a)["lincome"]
se_beta2 <- coef(summary(mod5_a))["lincome", "Std. Error"]
t_beta2 <- (beta2_hat - 1) / se_beta2
pval_beta2 <- pt(t_beta2, df = df.residual(mod5_a))

beta2_hat
```

```
##   lincome
## 0.1369553
```

```r
se_beta2
```

```
## [1] 0.02675545
```

```r
t_beta2
```

```
##   lincome
## -32.25679
```

```r
pval_beta2
```

```
##
## 0.000000000
```

```
##
## lhseval                      0.1213***
##                              (0.0177)
##
## Constant                    -0.8415***
##                              (0.2924)
##
## ------------------------------------------------
## Observations                    401
## R2                             0.1839
## Adjusted R2                    0.1757
## Residual Std. Error      0.0770 (df = 396)
## F Statistic          22.3131*** (df = 4; 396)
## ================================================
## Note:                *p<0.1; **p<0.05; ***p<0.01
```

```r
coef(mod5_e)
```

```
## (Intercept)     prpblck     lincome      prppov     lhseval
## -0.84151492  0.09755017 -0.05299043  0.05212289  0.12130565
```

Com a nova variavel, a equacao estimada fica:

$$\widehat{\log(psoda)} = -0.842 + 0.0976\, prpblck - 0.0530\, lincome + 0.0521\, prppov + 0.1213\, lhseval$$

O coeficiente de `lhseval` e aproximadamente `0.1213`. Como temos um modelo log-log nessa variavel, ele e uma elasticidade: um aumento de 1% no valor das casas da regiao esta associado a um aumento de aproximadamente `0.12\%` no preco do refrigerante, ceteris paribus.

Agora vamos observar o que acontece com a significancia individual de `lincome` e `prppov`.

```r
coef(summary(mod5_e))[c("lincome", "prppov", "lhseval"), c("Estimate", "Std. Error", "Pr(>|t|)")]
```

```
##            Estimate Std. Error            Pr(>|t|)
## lincome -0.05299043 0.03752608 0.15870700131425639
## prppov   0.05212289 0.13449922 0.69856986191565551
## lhseval  0.12130565 0.01768407 0.00000000002668167
```

Depois da inclusao de `lhseval`, tanto `lincome` quanto `prppov` deixam de ser individualmente significantes aos niveis usuais:

- p-valor de `lincome`: 0.1587;
- p-valor de `prppov`: 0.6986.

Ja `lhseval` permanece fortemente significante.

Por fim, vamos testar se `lincome` e `prppov` sao conjuntamente significativas.

```r
mod5_e_restrito <- lm(lpsoda ~ prpblck + lhseval, data = discrim)
anova(mod5_e_restrito, mod5_e)
```

```
## Analysis of Variance Table
##
## Model 1: lpsoda ~ prpblck + lhseval
## Model 2: lpsoda ~ prpblck + lincome + prppov + lhseval
##   Res.Df    RSS Df Sum of Sq      F  Pr(>F)
## 1    398 2.3911
## 2    396 2.3493  2  0.041797 3.5227 0.03045 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

O p-valor do teste F conjunto e aproximadamente 0.03045. Portanto, rejeitamos a hipotese nula de que os coeficientes de `lincome` e `prppov` sejam ambos zero ao nivel de 5%.

Em resumo:

- `lincome` e `prppov` nao sao individualmente significantes depois da inclusao de `lhseval`;
- mas elas sao **conjuntamente significativas** a 5%.